Timing-Only Channel Estimation and Equalization for OFDM Through a Dispersive Time-Modulated RIS

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***Abstract*—A time-modulated reconfigurable intelligent surface (TM-RIS) switches its elements several times within each OFDM symbol. The switching couples every received subcarrier to many transmitted ones, and with frequency-dispersive elements the coupling matrix is neither diagonal nor Toeplitz. Existing TM-RIS designs treat this coupling as interference, which caps the legitimate user's signal-to-interference ratio at -4.8 dB for a 75% duty cycle, and the receiver for the dispersive model was left open. We show that grouping the harmonic expansion by element state separates the coupling exactly into Toeplitz factors fixed by the switching timing, which the receiver knows, and per-state input responses that absorb the propagation *and* the element dispersion, which it does not. From this separation we build, to our knowledge, the first pilot-based receiver for this channel: a low-rank estimator that needs no element model, a blind estimator of the RIS clock offset that rests on the fact that an offset is exactly an opposite delay of the two hops, and an exact matrix-free equalizer. In pre-registered tests with 30 channels per point, the receiver comes within 0.1 bit/s/Hz of perfect channel knowledge with two pilot symbols on a shared-path channel and needs four to eight on 3GPP TDL-C. It beats every baseline we could build on every channel and pilot count, including an ideal-element estimator with the same prior and basis-expansion estimators with up to all harmonics and 32 pilot symbols. It decodes coded blocks within 0.14 dB of perfect knowledge, and it absorbs physical clock offsets with no significant rate loss.**

***Index Terms*—Reconfigurable intelligent surface, time modulation, OFDM, intercarrier interference, channel estimation, equalization.**

I. Introduction

A time-modulated RIS (TM-RIS), or space-time-coding metasurface, switches each element several times within an OFDM symbol [1]. The reflected signal then carries harmonics of the switching rate, which has been used for waveform security and directional modulation [2], [3]. For the user the surface serves, the same harmonics move energy from transmitted subcarrier m to received subcarrier $m + h$, so the channel matrix is not diagonal. Iudice *et al.* [4] recently derived a physically consistent model of this channel with frequency-dispersive (finite-memory) elements, as real varactor cells are. They showed that the coupling survives any cyclic prefix, that with dispersive elements the coupling matrix is not even Toeplitz. They assume the channel known and name joint subcarrier equalization as future work [4, Sec. V]; channel estimation is not treated. Prior TM-RIS designs keep a one-tap receiver for the legitimate user and treat the coupling as interference [2], [3]. A blind eavesdropper attack [5] relies on the Toeplitz structure of ideal, frequency-flat switching, which dispersion breaks. Generic estimators of doubly selective OFDM channels, such as the basis-expansion model (CE-BEM) [6], do not know where the coupling comes from: they must estimate every harmonic. Because rectangular switching leaves a $1/h^2$ energy tail, they either truncate the harmonics, which caps their accuracy, or keep all of them, which needs many pilots.

Novelty. The receiver of a TM-RIS link knows *how* the channel varies inside the symbol: the variation is the switching pattern, which the network sets. It does not know the propagation or the exact element response. We turn this observation into an exact structural statement and build the whole receiver on it. Our contributions are:

1) *Timing–propagation separation* (Proposition 1). Grouping the expansion of [4, eq. (5)] by element state gives $\boldsymbol{H} = \sum_{q,s} \boldsymbol{T}_{q,s} \circ (\boldsymbol{a}_q^{\mathrm{d}} \boldsymbol{v}_{q,s}^{\mathsf{T}})$. The Toeplitz matrices $\boldsymbol{T}_{q,s}$ depend only on the switching timing. The element dispersion enters only the per-state input response $\boldsymbol{v}_{q,s}$, together with the unknown BS–RIS channel. The regrouping is elementary and we credit it to the anchor model; what is new is what it implies for a receiver.
2) *A receiver that needs no element model*: a low-rank, shared-subspace estimator of $\boldsymbol{H}$ that treats $\boldsymbol{v}_{q,s}$ as unknown (Algorithm 1). Unlike [5], it does not rely on Toeplitz structure; unlike CE-BEM, it never estimates harmonics separately.
3) *Blind clock-offset recovery.* We prove that an offset between the RIS and the receiver is exactly an opposite delay of the two hops (Proposition 2), and we identify the offset physically (Proposition 3). This explains why a wrong offset overfits the pilots, and it leads to an estimator of the offset and the delay reference by Bayesian evidence (Algorithm 2).
4) *A pre-registered evaluation*: pass/fail criteria fixed before each run, 30 channels per point, paired confidence intervals, settings chosen on channels disjoint from the reported ones, the strongest baselines we could build, and failures reported.

We adopt and do not claim: the DPSS delay basis [7], [8], low-rank RIS channel models [9], matrix-free iterative equal-

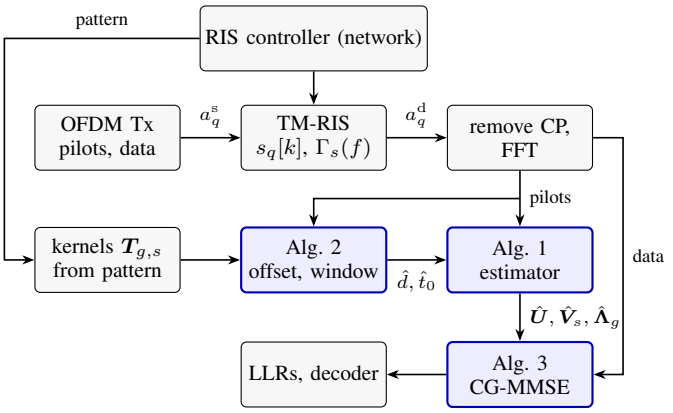


Fig. 1. System and receiver. Highlighted blocks are the proposed algorithms. The receiver uses the switching pattern, never the element model, and never forms the $M \times M$ coupling matrix.

ization [10] (exact here rather than a BEM approximation), and evidence-based model selection [11]. The closest timing work, blind offset estimation for RIS symbiotic radio [12], estimates the offset of a backscattered data stream from virtual subcarriers; our receiver instead uses the known pattern and needs no idle subcarriers.

Notation: $\circ$ and $\odot$ denote Hadamard products of matrices and vectors; $(\cdot)^{\mathsf{T}}$, $(\cdot)^{\mathsf{H}}$ and $(\cdot)^*$ denote transpose, conjugate transpose and conjugate; $\operatorname{sinc} x = \sin(\pi x)/(\pi x)$; $\mathcal{CN}$ denotes the circular complex Gaussian distribution.

II. System Model

A base station (BS) transmits OFDM with M subcarriers $f_m = f_c + m\Delta f$, $m \in \mathcal{M} = \{-M/2, \dots, M/2-1\}$, useful duration $T_u = 1/\Delta f$ and cyclic prefix (CP) T_{cp}, to a user through a RIS of Q elements (Fig. 1). Element q follows a periodic pattern $s_q[k] \in \{1, 2\}$, $k = 0, \dots, K-1$, of period T_u: it is in state $s_q[k]$ during slot $[kT_u/K, (k+1)T_u/K)$. The duty cycle D is the fraction of slots in state 1. In state s the element reflects with the single-resonance response of [4],

$$\Gamma_s(f) = -1 + \frac{2\xi_{r,s}}{j2\pi(f - f_{0,s}) + \xi_{r,s} + \xi_{i,s}}, \tag{1}$$

which for the published OpenRIS cell ($f_{0,1} = 3.471\,\mathrm{GHz}$, $f_{0,2} = 3.710\,\mathrm{GHz}$, $\xi_{r,s}/2\pi = 97.5$, $128.4\,\mathrm{MHz}$, $\xi_{i,s}/2\pi = 8.9$, $9.0\,\mathrm{MHz}$ [4, Tab. I]) drifts by about 47° in phase across a $100\,\mathrm{MHz}$ carrier at $f_c = 3.594\,\mathrm{GHz}$ (Fig. 2a). It is far from the frequency-flat ± 1 switch of earlier work. Let $a_q^{\mathrm{s}}(f)$ and $a_q^{\mathrm{d}}(f)$ be the BS–element and element–user responses. When the CP covers the cascaded delay spread plus the element memory (13.8 ns), the received frequency-domain symbol is [4]

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{w}, \quad H_{\bar{m}m} = \sum_{q=1}^{Q} a_q^{\mathrm{d}}(f_{\bar{m}})\, b_q^{[\bar{m}-m]}(f_m)\, a_q^{\mathrm{s}}(f_m), \tag{2}$$

where $\boldsymbol{w} \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2\boldsymbol{I})$ and $b_q^{[h]}(f)$ is the h-th Fourier coefficient, over one period, of the switched response of element q at input frequency f. Equation (2) is the ground truth of our evaluation. Its implementation agrees with an independent time-domain simulation (filter, gate, FFT, with no harmonic algebra) to 0.27% median relative error.

TABLE I
Assumptions, justification, and where they are tested.

Assumption	Justification; test of its violation
A1: pattern $s_q[k]$ known	The network drives the RIS controller, so the pattern is control information, like a pilot configuration. Without it the coupling cannot be told apart from propagation: a receiver that ignores one of two panels collapses to 1.15 bit/s/Hz (Sec. V-F).
A2: generalized CP	Condition of [4]. Checked on the cascade: 3GPP TDL at 30 and 100 ns is valid (≤ -56 dB of energy beyond the CP); 300 ns is not and is out of scope.
A3: period T_u, aligned up to an offset d	TM-IRS designs synchronize the pattern to the symbol [2]; the residual offset is estimated (Sec. IV-C).
A4: shared paths	A compact surface sees the same paths; this is what makes the factors low-rank. Tested with 3GPP TDL taps from random directions, spatially consistent CDL, 1024 elements and a near-field user.
A5: channel constant over a refresh block	The coherence time $0.423/f_D$ [13] spans 1188, 119 and 30 symbols at 3, 30 and 120 km/h. Mobility is tested with the refresh interval selected under pilot overhead (Sec. V-E).
A6: σ^2 known; no mutual coupling	σ^2 is estimated from empty resource elements. Mutual coupling is ignored, as in [4]; this is a stated limitation.

Assumptions. Table I lists each assumption with its justification and where its violation is evaluated. The receiver does *not* assume (1): it only defines the truth.

III. Structure of the Coupling

Proposition 1 (Timing–propagation separation)**.** *For a K-slot pattern of period T_u, $b_q^{[h]}(f) = \sum_{s=1}^{2} t_{q,s}[h]\Gamma_s(f)$ with*

$$t_{q,s}[h] = \frac{e^{-j\pi h/K}}{K} \operatorname{sinc}\left(\frac{h}{K}\right) \sum_{k:\, s_q[k]=s} e^{-j2\pi hk/K}. \tag{3}$$

Hence, with $[\boldsymbol{T}_{q,s}]_{\bar{m}m} = t_{q,s}[\bar{m} - m]$ (Toeplitz), $\boldsymbol{a}_q^{\mathrm{d}} = [a_q^{\mathrm{d}}(f_m)]_m$ and $\boldsymbol{v}_{q,s} = \boldsymbol{\Gamma}_s \odot \boldsymbol{a}_q^{\mathrm{s}}$,

$$\boldsymbol{H} = \sum_{q=1}^{Q} \sum_{s=1}^{2} \boldsymbol{T}_{q,s} \circ \left(\boldsymbol{a}_q^{\mathrm{d}} \boldsymbol{v}_{q,s}^{\mathsf{T}}\right). \tag{4}$$

Proof. In slot k the element applies $\Gamma_{s_q[k]}(f)$, so its switched response is $\sum_s \Gamma_s(f) \mathbb{1}_s(t)$, where $\mathbb{1}_s$ indicates the slots in state s. The h-th Fourier coefficient of the indicator of slot k is $\frac{1}{T_u}\int_{kT_u/K}^{(k+1)T_u/K} e^{-j2\pi ht/T_u} dt = \frac{1}{K} e^{-j\pi h/K} \operatorname{sinc}(h/K) e^{-j2\pi hk/K}$. Summing over the slots in state s gives (3). Substituting into (2), $t_{q,s}$ depends on $\bar{m} - m$ only, which gives (4). □

Proposition 1 has five consequences, which together define the receiver.

Corollary 1 (No element model)**.** *The element response enters* (4) *only through $\boldsymbol{v}_{q,s}$, which also contains the unknown BS–RIS channel. Estimating $\boldsymbol{v}_{q,s}$ as an unknown absorbs the*

dispersion exactly; the receiver needs only the timing kernels $\boldsymbol{T}_{q,s}$.

Corollary 2 (One-tap ceiling)**.** *For ideal* ± 1 *elements with a common pattern of duty* D, $b^{[0]} = t_1[0] - t_2[0] = 2D - 1$, *and Parseval's theorem gives* $\sum_h |b^{[h]}|^2 = \frac{1}{T_u}\int_0^{T_u} |\pm 1|^2 dt = 1$. *A one-tap receiver collects the fraction* $(2D-1)^2$ *of the reflected energy and sees the rest as interference, so its SIR is at most* $(2D-1)^2/(1-(2D-1)^2)$*:* $-4.8\,dB$ *at* $D = 0.75$. *Joint equalization collects all of it.*

Corollary 3 (Harmonic tail)**.** *From* (3), $|t_{q,s}[h]| \le |\operatorname{sinc}(h/K)|$, *which decays as* $1/|h|$, *so the energy outside* $|h| \le H$ *decays as* $1/H$. *For* $K = 8$ *and* $D = 0.75$ *it is 3.4%, 1.0% and 0.32% for* $H = 12$, *40 and 128 (Fig. 2b). A model truncated to* $|h| \le 12$ *therefore has a normalized error floor of* $10\log_{10} 0.034 = -14.7\,dB$, *whatever its estimator.*

Corollary 4 (Fast products)**.** $(\boldsymbol{T} \circ (\boldsymbol{a}\boldsymbol{b}^{\mathsf{T}}))\boldsymbol{x} = \boldsymbol{a} \odot (\boldsymbol{T}(\boldsymbol{b} \odot \boldsymbol{x}))$, *and* $\boldsymbol{T}\boldsymbol{z}$ *is a linear convolution. Hence* $\boldsymbol{H}\boldsymbol{x}$ *and* $\boldsymbol{H}^{\mathsf{H}}\boldsymbol{y}$ *cost* $\mathcal{O}(QM \log M)$ *with FFTs of length at least* $3M - 2$, *and* $\boldsymbol{H}$ *is never formed.*

Proposition 2 (An offset is an opposite delay)**.** *If the switching runs* d *slots late relative to the receiver's timing, each kernel becomes* $\boldsymbol{D}\boldsymbol{T}\boldsymbol{D}^*$ *with* $\boldsymbol{D} = \operatorname{diag}(e^{-j2\pi md/K})_{m\in\mathcal{M}}$, *and*

$$(\boldsymbol{D}\boldsymbol{T}\boldsymbol{D}^*) \circ (\boldsymbol{a}\boldsymbol{b}^{\mathsf{T}}) = \boldsymbol{T} \circ \big((\boldsymbol{D}\boldsymbol{a})(\boldsymbol{D}^*\boldsymbol{b})^{\mathsf{T}}\big). \tag{5}$$

Multiplying a frequency response by $\boldsymbol{D}$ *delays it by* dT_u/K; *multiplying it by* $\boldsymbol{D}^*$ *advances it by the same amount.*

Proof. A shift of d slots multiplies the h-th coefficient by $e^{-j2\pi hd/K} = e^{-j2\pi \bar{m}d/K} e^{j2\pi md/K}$ for $h = \bar{m} - m$. This is (5) entrywise. □

Proposition 3 (Physical timing)**.** *Let the two hops have bulk delays* τ_b *and* τ_u, *and let the receiver synchronize to the first cascaded arrival. (i) If the RIS controller is locked to the signal arriving at the RIS, the synchronized* $\boldsymbol{H}$ *equals the one without bulk delays. (ii) If it is locked to the BS clock, the synchronized* $\boldsymbol{H}$ *equals the one without bulk delays and with offset* $d = -\tau_b K/T_u$.

Proof. The delays multiply the input side by $e^{-j2\pi f_m \tau_b}$ and the output side by $e^{-j2\pi f_{\bar{m}}\tau_u}$; synchronization multiplies the output by $e^{j2\pi f_{\bar{m}}(\tau_b+\tau_u)}$. In (i) the switching edges are delayed by τ_b, which by Proposition 2 contributes $e^{-j2\pi(\bar{m}-m)\Delta f \tau_b}$, and all phases cancel. In (ii) the edges are not delayed, and what remains is the phase of an offset of $-\tau_b$. □

Both identities hold numerically to $7 \cdot 10^{-15}$. A practical RIS therefore sees either no offset or an offset equal to its BS distance, tens to hundreds of nanoseconds; the latter costs 1.6/4.2 bit/s/Hz at 20/30 dB if ignored (Table III).

IV. The Receiver

A. Low-rank model with a delay prior

Under A4, the user-hop response of element q is $a_q^{\mathrm{d}}(f) = \sum_\ell g_\ell e^{-j\boldsymbol{k}\boldsymbol{u}_\ell^{\mathsf{T}}\boldsymbol{r}_q} e^{-j2\pi f\tau_\ell}$, so $\boldsymbol{a}_q^{\mathrm{d}} = \boldsymbol{E}\boldsymbol{\alpha}_q$ with $\boldsymbol{E} = [e^{-j2\pi f_m \tau_\ell}]_{m,\ell}$ common to all elements. Likewise $\boldsymbol{v}_{q,s} = \operatorname{diag}(\boldsymbol{\Gamma}_s)\boldsymbol{E}'\boldsymbol{\beta}_q$, where $\boldsymbol{\beta}_q$ (the input-hop path gains with the element's geometric phases) is common to both states. Hence $[\boldsymbol{a}_1^{\mathrm{d}}, \ldots, \boldsymbol{a}_Q^{\mathrm{d}}]$ has rank at most the number of paths, whatever Q. Every hop response is delay-limited to a window $[t_0, t_0 + \tau_{\max}]$, where $\tau_{\max}$ also covers the element memory. We represent the responses with the first L discrete prolate spheroidal sequences (DPSS) [7], [8]. These are the columns of $\boldsymbol{F} \in \mathbb{C}^{M\times L}$: the leading eigenvectors of $R_{nm} = e^{-j\pi(n-m)\Delta f\tau_{\max}} \operatorname{sinc}((n-m)\Delta f\tau_{\max})$, the frequency correlation of a response with delays uniform in the window. Their normalized eigenvalues $\boldsymbol{\lambda}$ serve as prior variances. Compressing the path gains to rank r gives $\boldsymbol{a}_q^{\mathrm{d}} = \boldsymbol{F}\boldsymbol{U}\tilde{\boldsymbol{\alpha}}_q$ and $\boldsymbol{v}_{q,s} = \boldsymbol{F}\boldsymbol{V}_s\tilde{\boldsymbol{\beta}}_q$. The second relation holds because Γ_s is smooth (memory 13.8 ns): $\operatorname{diag}(\boldsymbol{\Gamma}_s)\boldsymbol{E}'$ stays delay-limited, so it lies in the same DPSS span with a state-specific coefficient matrix $\boldsymbol{V}_s$, while $\tilde{\boldsymbol{\beta}}_q$ is shared by the two states. With G distinct patterns (groups $\mathcal{G}_g$) and rank r, (4) becomes

$$\boldsymbol{H} = \sum_{g=1}^{G}\sum_{s=1}^{2} \boldsymbol{T}_{g,s} \circ \big((\boldsymbol{F}\boldsymbol{U})\boldsymbol{\Lambda}_g(\boldsymbol{F}\boldsymbol{V}_s)^{\mathsf{T}}\big), \tag{6}$$

with $\boldsymbol{U}, \boldsymbol{V}_s \in \mathbb{C}^{L\times r}$ and $\boldsymbol{\Lambda}_g = \sum_{q\in\mathcal{G}_g} \tilde{\boldsymbol{\alpha}}_q\tilde{\boldsymbol{\beta}}_q^{\mathsf{T}} \in \mathbb{C}^{r\times r}$. The number of unknowns falls from up to M^2 to $3Lr + Gr^2$. An unswitched direct path is the special case of equal input responses in both states (since $\boldsymbol{T}_{q,1} + \boldsymbol{T}_{q,2} = \boldsymbol{I}$), so (6) contains it without change.

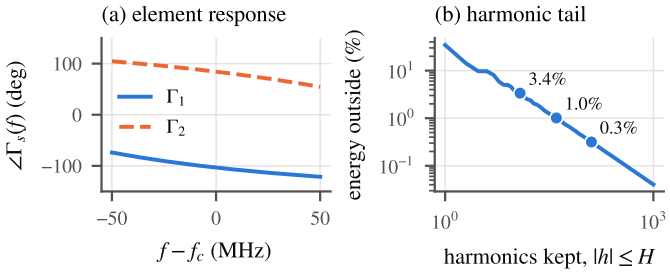


Fig. 2. (a) Phase of the two states of the OpenRIS cell over the 100 MHz carrier. (b) Fraction of the coupling energy outside $|h| \le H$ for an ideal switch, $K = 8$, $D = 0.75$ (Corollary 3).

B. Timing-only estimation (Algorithm 1)

Let pilot symbol p be $\boldsymbol{s}_p$, with received $\boldsymbol{y}_p$, and let $\boldsymbol{G}_{g,s,p} = \boldsymbol{T}_{g,s}\operatorname{diag}(\boldsymbol{s}_p)\boldsymbol{F}$, computed once by FFT. By Corollary 4,

$$\boldsymbol{H}\boldsymbol{s}_p = \sum_{g,s} \big[(\boldsymbol{F}\boldsymbol{U}\boldsymbol{\Lambda}_g) \circ (\boldsymbol{G}_{g,s,p}\boldsymbol{V}_s)\big]\mathbf{1}_r, \tag{7}$$

which is linear in $\boldsymbol{U}$, in $[\boldsymbol{V}_1; \boldsymbol{V}_2]$ and in $\{\boldsymbol{\Lambda}_g\}$ when the other two are fixed. For each block $\boldsymbol{\theta}$ the N_p stacked pilots give $\boldsymbol{y} = \boldsymbol{A}_{\boldsymbol{\theta}}\boldsymbol{\theta} + \boldsymbol{w}$. With the prior $\boldsymbol{\theta} \sim \mathcal{CN}(\mathbf{0}, \boldsymbol{P})$, $\boldsymbol{P} = \operatorname{diag}(\boldsymbol{p})$, the columns of $\boldsymbol{U}$ and $\boldsymbol{V}_s$ carry variances $\varsigma\boldsymbol{\lambda}$, with the scale ς set by moment matching on $\|\boldsymbol{y}\|^2$. The LMMSE update

$$\hat{\boldsymbol{\theta}} = \boldsymbol{P}^{1/2}\big(\boldsymbol{P}^{1/2}\boldsymbol{A}_{\boldsymbol{\theta}}^{\mathsf{H}}\boldsymbol{A}_{\boldsymbol{\theta}}\boldsymbol{P}^{1/2} + \sigma^2\boldsymbol{I}\big)^{-1}\boldsymbol{P}^{1/2}\boldsymbol{A}_{\boldsymbol{\theta}}^{\mathsf{H}}\boldsymbol{y} \tag{8}$$

stays well posed when some prior variances are tiny. Alternating (8) over the blocks is non-convex, so the initialization

Algorithm 1 Timing-only low-rank estimation

Input: pilots $\{\boldsymbol{y}_p, \boldsymbol{s}_p\}$; kernels $\boldsymbol{T}_{g,s}$; $\boldsymbol{F}, \boldsymbol{\lambda}$; σ^2; rank r; iterations I

$\boldsymbol{G}_{g,s,p} \leftarrow \boldsymbol{T}_{g,s}\,\mathrm{diag}(\boldsymbol{s}_p)\boldsymbol{F}$ by FFT

$[\hat{\boldsymbol{C}}_1, \hat{\boldsymbol{C}}_2] \leftarrow$ LMMSE of the linear model; rank-r SVD $\rightarrow \boldsymbol{U} = \boldsymbol{Y}_r \boldsymbol{\Sigma}_r^{1/2}$, $[\boldsymbol{V}_1; \boldsymbol{V}_2] = \boldsymbol{W}_r^* \boldsymbol{\Sigma}_r^{1/2}$, $\boldsymbol{\Lambda}_g = \boldsymbol{I}$

for $i = 1, \ldots, I$ **do**

  $\{\boldsymbol{\Lambda}_g\} \leftarrow$ ridge least squares on (7); $\boldsymbol{U} \leftarrow$ (8); $[\boldsymbol{V}_1; \boldsymbol{V}_2] \leftarrow$ (8)

Output: factors of (6) ($\boldsymbol{H}$ is not formed)

matters. Ignoring the coupling between groups gives a *linear* model, $\boldsymbol{H} \approx \sum_s \bar{\boldsymbol{T}}_s \circ (\boldsymbol{F}\boldsymbol{C}_s\boldsymbol{F}^{\mathsf{T}})$ with $\bar{\boldsymbol{T}}_s = \sum_g \boldsymbol{T}_{g,s}$, which is exact for $G = 1$. Its LMMSE estimate with prior $\varsigma(\boldsymbol{\lambda} \otimes \boldsymbol{\lambda})$, followed by a rank-$r$ SVD $[\hat{\boldsymbol{C}}_1\ \hat{\boldsymbol{C}}_2] \approx \boldsymbol{Y}\boldsymbol{\Sigma}\boldsymbol{W}^{\mathsf{H}}$, initializes the factors. From this initialization ALS converges in one or two iterations; random initializations end 3–5 times worse.

Why two pilot symbols can suffice. At $M = 256$ with $L = 16$ and $r = 8$ there are 448 unknowns against 512 observations from two pilot symbols, yet the problem is well determined because the prior concentrates each hop on about $M\Delta f \tau_{\max} + 1$ dimensions. We measured the effective degrees of freedom of the linear stage, the trace of its LMMSE hat matrix: 19–27 at 20 dB and 23–33 at 30 dB for 1–8 pilot symbols, against 256–2048 observations. With two pilot symbols the achieved coupling-matrix error is 5–6 dB above the Bayesian bound under the assumed prior (-26.4 vs -32.4 dB at 20 dB). The gap is not estimator suboptimality: real channels do not follow the uniform-delay prior. An earlier version kept the linear estimate whenever it fitted the pilots better. On fresh 3GPP channels this rule was an overfitting trap (chosen in 37–57% of channels at 30 dB, worse rate), so Algorithm 1 always returns the low-rank estimate.

C. Blind clock offset and delay reference (Algorithm 2)

By Proposition 2 a candidate offset d replaces each kernel by $\boldsymbol{D}_d \boldsymbol{T}_{g,s} \boldsymbol{D}_d^*$, and a window start t_0 replaces $\boldsymbol{F}$ by $\boldsymbol{F}_{t_0} = \mathrm{diag}(e^{-j2\pi m \Delta f t_0})\boldsymbol{F}$. For each candidate the linear stage is a Gaussian model, $\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{c} + \boldsymbol{w}$ with $\boldsymbol{c} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{P})$, and its log-evidence is, up to a constant,

$$\mathcal{E}(d, t_0) = -\boldsymbol{y}^{\mathsf{H}}\boldsymbol{C}^{-1}\boldsymbol{y} - \log\det \boldsymbol{C}, \qquad \boldsymbol{C} = \boldsymbol{\Phi}\boldsymbol{P}\boldsymbol{\Phi}^{\mathsf{H}} + \sigma^2\boldsymbol{I}. \tag{9}$$

With the Cholesky factor $\boldsymbol{R}^{\mathsf{H}}\boldsymbol{R} = \sigma^{-2}\boldsymbol{P}^{1/2}\boldsymbol{\Phi}^{\mathsf{H}}\boldsymbol{\Phi}\boldsymbol{P}^{1/2} + \boldsymbol{I}$ and the Woodbury identity, $\log\det \boldsymbol{C} = MN_p \log\sigma^2 + 2\sum_i \log R_{ii}$ and $\boldsymbol{y}^{\mathsf{H}}\boldsymbol{C}^{-1}\boldsymbol{y} = \sigma^{-2}\|\boldsymbol{y}\|^2 - \sigma^{-4}\|\boldsymbol{R}^{-\mathsf{H}}\boldsymbol{P}^{1/2}\boldsymbol{\Phi}^{\mathsf{H}}\boldsymbol{y}\|^2$, so the evidence costs one $2L^2$-dimensional factorization per candidate.

Why the evidence. By (5), a wrong offset can still explain the pilots, through a delayed user hop and an advanced BS hop. The pilot residual is therefore nearly flat in d for physical offsets below about 100 ns. For such offsets, the residual of the advanced response after projection onto the delay basis is only $2.7 \cdot 10^{-4}$ at 100 ns (relative). A wrong offset thus

Algorithm 2 Offset and delay reference by evidence

Input: pilots; nominal kernels; $\boldsymbol{F}, \boldsymbol{\lambda}$; σ^2; $\mathcal{T}_0 = \{-150, -75, 0, 75\}$ ns

for $t_0 \in \mathcal{T}_0$ **do**

  $d_0 \leftarrow \arg\max_d \mathcal{E}(d, t_0)$ on a 41-point grid over $[-\frac{1}{2}, \frac{1}{2}]$ slot

  $\hat{d}_{t_0} \leftarrow$ golden-section search of $\mathcal{E}(\cdot, t_0)$ on $d_0 \pm 0.025$ (14 steps)

Output: $(\hat{d}, \hat{t}_0) = \arg\max_{t_0} \mathcal{E}(\hat{d}_{t_0}, t_0)$; run Algorithm 1 with them

Algorithm 3 Matrix-free CG-MMSE equalization

Input: data $\boldsymbol{Y}_d$ (symbols as columns); factors of $\hat{\boldsymbol{H}}$; σ^2; $\epsilon = 3 \cdot 10^{-3}$; $i_{\max} = 50$

$\boldsymbol{B} \leftarrow \hat{\boldsymbol{H}}^{\mathsf{H}}\boldsymbol{Y}_d$; $\boldsymbol{X} \leftarrow \boldsymbol{0}$; $\boldsymbol{R} \leftarrow \boldsymbol{B}$; $\boldsymbol{P} \leftarrow \boldsymbol{R}$

for $i = 1, \ldots, i_{\max}$ **do** ▷ per column

  $\boldsymbol{Q} \leftarrow \hat{\boldsymbol{H}}^{\mathsf{H}}(\hat{\boldsymbol{H}}\boldsymbol{P}) + \sigma^2\boldsymbol{P}$ ▷ batched FFT products

  $\alpha \leftarrow \|\boldsymbol{r}\|^2 / \boldsymbol{p}^{\mathsf{H}}\boldsymbol{q}$; $\boldsymbol{x} \leftarrow \boldsymbol{x} + \alpha\boldsymbol{p}$; $\boldsymbol{r}' \leftarrow \boldsymbol{r} - \alpha\boldsymbol{q}$

  if $\|\boldsymbol{r}'\|/\|\boldsymbol{b}\| < \epsilon$ for all columns **break**

  $\boldsymbol{p} \leftarrow \boldsymbol{r}' + (\|\boldsymbol{r}'\|^2/\|\boldsymbol{r}\|^2)\boldsymbol{p}$; $\boldsymbol{r} \leftarrow \boldsymbol{r}'$

overfits: it fits through improbable shifted responses. The log-determinant in (9) penalizes exactly this. With two held-out pilot symbols, the offset that minimizes the held-out residual agrees with the evidence maximizer within 0.005 slot in 93% of channels. The window start must be estimated too. After synchronization the first arrivals of the two hops sum to zero up to the offset, so a window that starts after the true reference cannot hold both hops; a synchronization error has the same effect. The evidence of different priors is directly comparable, so t_0 is chosen jointly with d.

D. Matrix-free equalization and complexity

The MMSE symbol estimate solves $(\hat{\boldsymbol{H}}^{\mathsf{H}}\hat{\boldsymbol{H}} + \sigma^2\boldsymbol{I})\boldsymbol{x} = \hat{\boldsymbol{H}}^{\mathsf{H}}\boldsymbol{y}$. We solve it by conjugate gradients (CG), as in [10]. Each product goes through (6): for $G = 1$, the FFTs of all r columns and all data symbols form one batched call per state, the two states are summed in the frequency domain, and one batched inverse FFT follows (Algorithm 3). Unlike the BEM approximation in [10], this representation is exact. The stopping tolerance was chosen by a pre-registered sweep. With $\epsilon = 10^{-2}$ the symbol error rate (SER) is 1.15 times the dense solve's at $M = 3276$; with $\epsilon = 3 \cdot 10^{-3}$ it is within 2% of it at every M from 1024 to 6552, after 8–20 iterations.

Complexity (complex multiply–adds). One ALS iteration costs $\sum_{n \in \{Lr, 2Lr, Gr^2\}} (MN_p n^2 + n^3/3)$; the linear stage costs $MN_p(2L^2)^2 + (2L^2)^3/3$. For $G = 1$, $r = 8$, $N_p = 2$ and 20 iterations this totals $5.7 \cdot 10^8$ at $M = 256$ and $3.9 \cdot 10^{10}$ at the full $M = 3276$. One CG iteration costs two products of $\mathcal{O}(rM \log M)$, together about $1.7 \cdot 10^7$ operations at $M = 3276$, so a symbol costs at most $3.4 \cdot 10^8$ with 20 iterations. The dense alternative forms $\hat{\boldsymbol{H}}$ (172 MB at $M = 3276$, 687 MB at $M = 6552$) and factors $\hat{\boldsymbol{H}}^{\mathsf{H}}\hat{\boldsymbol{H}} + \sigma^2\boldsymbol{I}$ once per estimate ($4.7 \cdot 10^{10}$ operations), after which each

symbol costs $\mathcal{O}(M^2)$. Measured on an idle 4-thread CPU ($\epsilon = 3\cdot10^{-3}$, $M = 3276$), dense equalization of 1, 12, 40 and 120 symbols per estimate took 2.51, 2.54, 2.58 and 2.78 s and CG 0.23, 2.51, 8.2 and 23.8 s, with CG's SER within 2% of dense. The break-even is thus about 12 symbols, not the 140 the operation count suggests, because the dense factorization runs as level-3 BLAS. CG is the faster choice for the 6–15-symbol blocks selected at 30 and 120 km/h (Sec. V-E); dense is 3–9× faster for slow channels. CG's memory is linear in M: at $M = 6552$ it equalized 120 symbols in 51 s without the 687 MB matrix. Warm-starting a refresh from the previous factors (two ALS iterations, no linear stage) took 1.4 s against 12.5 s from scratch, with the same SER.

V. Evaluation

Setup. We use the n78 carrier with $\Delta f = 30$ kHz, $T_{\mathrm{cp}} = 2.34\,\mu$s, $Q = 16$ elements (4×4, $\lambda/2$), $K = 8$, $D = 0.75$, identical patterns unless stated, and the OpenRIS cell (1). Comparisons use $M = 256$ (central 128 subcarriers scored); full-band results use $M = 3276$. Channels:

- a *shared-path* channel: per hop, one line-of-sight and five scattered paths, 30 ns rms, Rician factor 10 dB;
- 3GPP TDL-A/C/D [14] at 100 ns, with taps from random directions.

Pilots are QPSK pilot symbols. The estimator uses $\tau_{\max} = 0.3\,\mu$s, $L = 16$, $r = 8$ on the shared-path channel and $\tau_{\max} = 1\,\mu$s, $L = \lceil M\Delta f\tau_{\max}\rceil + 10$, $r = 16$ on TDL. These hyperparameters were frozen before the tests below. The metric is the achievable rate of the MMSE filter $\boldsymbol{W}$ built from the estimate and applied to the *true* channel: with $\boldsymbol{G} = \boldsymbol{W}\boldsymbol{H}$, $\mathrm{SINR}_m = |G_{mm}|^2/(\sum_{n\neq m}|G_{mn}|^2 + \sigma^2\|\boldsymbol{w}_m\|^2)$, and $R = \mathrm{mean}_m \log_2(1 + \mathrm{SINR}_m)$. The *loss* is R with perfect channel state information (CSI) minus R with the estimate. Each test was pre-registered with its pass criterion and uses 30 independent channels per point; intervals are paired 95% t-intervals.

Baselines. All baselines get the same pilots, the same prior family and, for fairness, three equalizers each: naive MMSE, robust MMSE with the estimation-error power taken from the pilot residual, and robust MMSE with the oracle error power. Figures show the residual-based robust MMSE, which is usable in practice; paired comparisons take, per channel, the best over all baselines and all three equalizers.

- *One-tap*: today's OFDM receiver.
- *Ideal-element LMMSE*: the strongest estimator that assumes the textbook element, $\boldsymbol{H} = (\boldsymbol{T}_1 - \boldsymbol{T}_2)\circ(\boldsymbol{F}\boldsymbol{C}\boldsymbol{F}^{\mathsf{T}})$ with our prior.
- *CE-BEM* [6]: $\boldsymbol{H} = \sum_{|h|\le H_b} \boldsymbol{S}_h\,\mathrm{diag}(\boldsymbol{F}_2\boldsymbol{c}_h)$ with up to all $2M-1$ harmonics, solved matrix-free (equal to the dense solver to $7\cdot10^{-9}$); with all harmonics it is run from eight pilot symbols.
- *OMP* [15] over a harmonic×delay dictionary.

A. Comparison with the strongest baselines

Fig. 3 gives the loss to perfect CSI. The proposed estimator is the only one that comes within 0.3 bit/s/Hz with at most 32 pilot symbols. It needs two on the shared-path channel (loss 0.07 ± 0.02 / 0.11 ± 0.03 at 20/30 dB) and four on TDL-C (0.19 at 20 dB; eight for 0.18 at 30 dB); the longer delay window of TDL means more unknowns. The one-tap receiver is capped by Corollary 2 (0.43 bit/s/Hz at both SNRs).

The strongest baseline is the ideal-element LMMSE. With eight or more pilot symbols it loses 0.4–0.5 bit/s/Hz at 20 dB but 1.9–2.2 at 30 dB, and more pilots do not help it. This is the signature of a *model* error: at high SNR the dispersion it ignores dominates. It is the quantitative case for Corollary 1. CE-BEM stalls when truncated (Corollary 3). With all harmonics it has 8687 unknowns at $M = 256$, and even with 32 pilot symbols it reaches only the ideal-element level (0.43 at 20 dB on the shared-path channel, 1.17 on TDL-C). Our receiver with two pilot symbols is therefore better than full CE-BEM with 32. Paired per channel against the best baseline on that channel (minimum over all baselines and equalizers), ours gains 0.41 ± 0.02 / 2.16 ± 0.04 bit/s/Hz at $N_p = 2$ on the shared-path channel and 0.30 ± 0.06 / 1.14 ± 0.25 on TDL-C. At every pilot count from 2 to 32 the gain is 0.3–0.4 at 20 dB and 0.5–2.2 at 30 dB, and every interval excludes zero. If the detector does not know G_{mm} and uses its estimate, every loss grows by at most 0.15. One pre-registered sanity check failed. Truncated CE-BEM still loses rate from 20 to 30 dB even with the oracle robust equalizer, because a scalar error power cannot describe an error concentrated on the missing harmonics.

TABLE II
Estimator unchanged on data violating its model: relative change of $\boldsymbol{H}$ and extra loss (bit/s/Hz, paired, 30 channels).

Violation	$\Delta\boldsymbol{H}$	20 dB	30 dB
Time-domain truth	0.006	-0.004 ± 0.005	$+0.006\pm0.009$
Switching transients	0.001	$+0.001\pm0.001$	$+0.003\pm0.006$
State-dep. output, $\epsilon = 0.2$	0.40	≤ 0.004	$+0.034$
Joint-state coupl., $\kappa = 0.08$	0.29	≤ 0.002	$+0.015$
Resonance spread 10 MHz	0.18	$+0.003\pm0.008$	$+0.046\pm0.044$
Beam squint, $Q = 256$	0.016	≤ 0.001	-0.002

B. No element model: robustness beyond the generating model

A receiver tested on data from its own model can look better than it is. Table II runs the estimator *unchanged* on data that violate its structure, paired with the same channel without the change. In every case the extra loss stays within 0.05 bit/s/Hz, which passes the pre-registered bar of 0.3. Element-to-element resonance spread of 5–20 MHz costs 0.06–0.15 bit/s/Hz. A six-parameter fit of the element model is not better than timing-only ($+0.08\pm0.05$ worse at 30 dB) and fails with element spread ($+0.73\pm0.43$). On a 16-element surface the linear first stage provides most of the gain (loss 0.10/0.23 at $N_p = 2$); the low-rank step and one decision-directed pass reduce it to 0.07/0.11 and 0.02/0.03. The low-rank property itself matters most for large surfaces: the effective rank of the true element-domain matrix (99% energy, CDL-C, 98 MHz, beam squint) is 5, 9, 13 and 16 for $Q = 16$, 64, 256 and 1024.

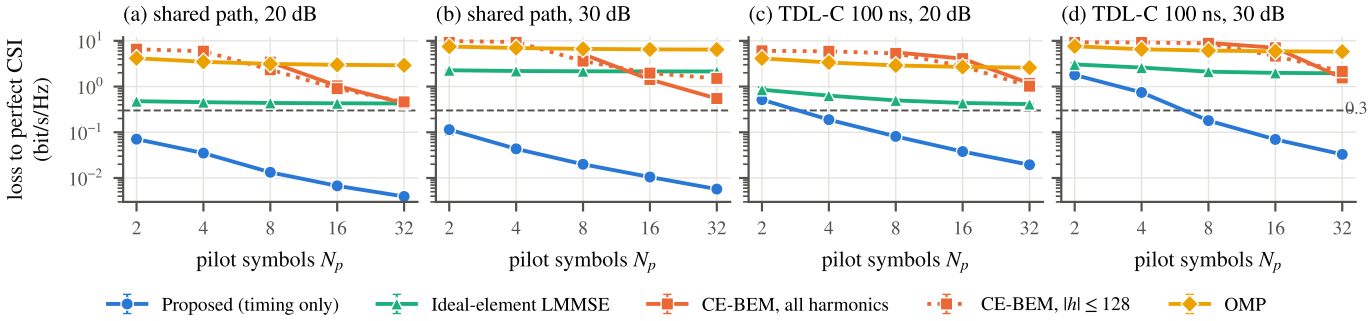


Fig. 3. Loss to perfect CSI versus pilot symbols (30 channels, 95% intervals, residual-based robust MMSE for every estimator). Dashed line: 0.3 bit/s/Hz. Perfect-CSI rates: 6.62/9.92 (shared path), 6.20/9.38 bit/s/Hz (TDL-C) at 20/30 dB.

TABLE III
TIMING: LOSSES AND PAIRED DIFFERENCES (BIT/S/HZ, 30 CHANNELS).

	20 dB	30 dB
BS-clock RIS, offset ignored	1.62	4.23
Alg. 2 minus known offset	$+0.011 \pm 0.019$	$+0.033 \pm 0.066$
Incident-locked: Alg. 2 minus none	$+0.001 \pm 0.007$	$+0.040 \pm 0.082$
Sync error ≤ 150 ns, fixed window	0.22	1.16
window chosen by evidence	0.05 ± 0.05	0.05 ± 0.08
$N_p = 1$: residual minus evidence	1.49 ± 0.56	2.83 ± 1.19

C. Timing

Table III tests Algorithm 2 with physical bulk delays ($\tau_b \sim \mathcal{U}(50, 300)$ ns, $\tau_u \sim \mathcal{U}(30, 300)$ ns). A BS-clock-locked RIS needs the search, and with the search the rate is within noise of known timing. An incident-signal-locked RIS needs no search, and running it anyway costs nothing significant. A synchronization error of up to ± 150 ns breaks a fixed delay window, but not the jointly chosen one. Is the evidence necessary? On a fine grid the pilot residual of the full low-rank estimate also finds the offset with two or more pilot symbols (not significantly worse). With a single pilot symbol it overfits: it is worse by 1.49 ± 0.56 / 2.83 ± 1.19 bit/s/Hz, while the evidence stays within 0.03 of known timing at 20 dB. The evidence is also 5–58 times cheaper per search (3.2 vs 17.2 s at $M = 256$; 11.7 vs 674 s on TDL-C), because it needs no low-rank estimate per candidate. Random per-edge jitter cannot be estimated. It must stay below about 4 ns rms at 30 dB (20 ns at 20 dB), whereas ordinary clocks have picosecond jitter.

D. Full band and coded link

At the full band ($M = 3276$, 30 channels) the SER with two pilot symbols is 1.10 [1.06, 1.15] times the perfect-CSI SER at 20 dB with 16-QAM and 1.37 [1.18, 1.86] times at 28 dB with 64-QAM (95% clustered-bootstrap intervals). One decision-directed pass brings these to 1.05 and 1.12. CE-BEM ($|h| \le 8$) is 18 and 32 times worse; the one-tap receiver's SER is 0.84 and 0.96. Fig. 4 shows the coded link: a rate-1/2 convolutional code, 16-QAM, max-log LLRs from each receiver's own estimate, and 2000 blocks per point. The proposed receiver crosses 10% block error rate (BLER) at 8.51 dB against 8.37 dB with perfect CSI, a gap of 0.14 dB [0.09, 0.20]. One-tap never decodes, and truncated CE-BEM floors at 34%.

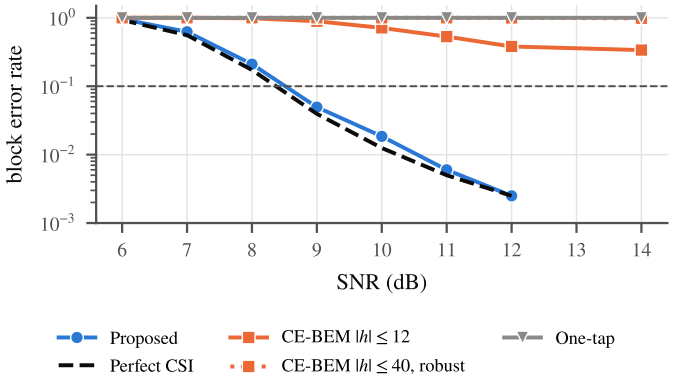


Fig. 4. Coded BLER, $N_p = 2$, 2000 blocks per point. Gap to perfect CSI at 10% BLER: 0.14 dB [0.09, 0.20].

E. Mobility and pilot overhead

When the user moves, only the RIS–user hop changes. Each path rotates by $e^{j2\pi\nu_\ell n T_s}$ between symbols and also mixes subcarriers within a symbol. That mixing is -60, -47 and -34 dB below the signal at 3, 30 and 120 km/h, so the main effect is the ageing of the estimate. With N_p pilot symbols every N symbols, the net rate is $R_{\text{net}} = \frac{1}{N}\sum_{n=N_p}^{N-1} R(n)$, where $R(n)$ uses the estimate on symbol n with one common phase corrected per symbol, as 5G phase-tracking pilots do. Table IV selects (N_p, N) and, for CE-BEM, the harmonic range on 20 channels, and reports them on 20 *other* channels. Ours beats the best CE-BEM in all 24 cases, by 1.0–3.9 bit/s/Hz, every interval above zero (CE-BEM selects 16–32 pilot symbols). Selecting on the reported channels changes the numbers by at most 0.13. Net of pilots, ours keeps 75–93% of the perfect-CSI rate on the shared path and 73–88% on TDL at 3 km/h, but only 46–66% on TDL at 120 km/h: with a 30-symbol coherence time the selected design spends one pilot symbol in every 6. Slow channels select two (shared) or four (TDL) pilot symbols per 40–120 symbols. High mobility is thus a limit of pilot-based estimation, not of the coupling model. Two decision-directed trackers (LMMSE and Kalman)

TABLE IV
NET THROUGHPUT AFTER PILOT OVERHEAD (BIT/S/HZ), (N_p, N) SELECTED ON 20 CHANNELS AND REPORTED ON 20 DISJOINT CHANNELS; OURS − CE-BEM PAIRED, 95% INTERVAL.

		20 dB			30 dB		
Channel	km/h	ours	BEM	diff.	ours	BEM	diff.
Shared	3	6.24	5.24	1.01 ± 0.14	8.99	7.34	1.65 ± 0.34
Shared	30	5.97	4.40	1.57 ± 0.30	8.56	5.42	3.14 ± 0.40
Shared	120	5.26	3.15	2.11 ± 0.30	7.30	3.38	3.92 ± 0.53
TDL-A	3	5.38	3.71	1.67 ± 0.17	7.37	4.88	2.49 ± 0.21
TDL-A	30	4.56	1.60	2.97 ± 0.13	5.51	1.68	3.84 ± 0.27
TDL-A	120	3.81	1.08	2.73 ± 0.28	4.35	1.07	3.27 ± 0.58
TDL-C	3	5.34	3.61	1.74 ± 0.16	7.23	4.81	2.41 ± 0.27
TDL-C	30	4.53	1.65	2.88 ± 0.29	5.53	1.81	3.72 ± 0.42
TDL-C	120	3.88	0.75	3.13 ± 0.31	4.61	0.74	3.88 ± 0.54
TDL-D	3	5.17	3.61	1.56 ± 0.17	6.80	4.75	2.05 ± 0.41
TDL-D	30	4.65	2.17	2.48 ± 0.25	5.85	2.39	3.46 ± 0.42
TDL-D	120	3.92	1.15	2.77 ± 0.31	4.68	1.15	3.53 ± 0.69

were worse than reusing the pilot estimate, because wrong decisions accumulate after about ten symbols.

F. Hardware, deployment and the cost of time modulation

Impairments. Residual carrier-frequency offset must stay below about 1% of the spacing (0.10/0.33 bit/s/Hz at 20/30 dB). Oscillator phase noise must stay below about 100 Hz linewidth or be tracked: at 100 Hz the loss over a genie that knows the channel but not the phase noise is 0.16/0.33. Switching rise times up to 42 ns need no calibration (≤ 0.011 at 30 dB). For longer ones, the kernels are built from a ramp whose rise time is calibrated or fitted blindly from the pilots, within 0.03 slot of the truth. Rise time also sets the emission. With ideal edges the adjacent-channel leakage ratio of the reflected signal is 36.8 dB, below the 45 dB required of a base station [16]; edges of 0.05 slot give 52.7 dB. They also shrink our lead over truncated estimators: at a ramp of 0.3 slot, CE-BEM with eight pilot symbols reaches 8.2 against our 8.56 bit/s/Hz with two.

Deployment. An unblocked direct path needs no change ($T_{q,1} + T_{q,2} = \boldsymbol{I}$): at equal power the rate is 5.28/8.13 against 5.39/8.49 bit/s/Hz with perfect CSI. Two panels with different patterns ($G = 2$) give 5.54/8.67 against 5.64/8.86. An unknown OFDM interferer at 0–20 dB SIR costs 0.03–0.17. Spatially consistent 3GPP CDL channels give the same loss as TDL within 0.07.

Cost of time modulation. At equal transmit power and 20 dB, a TM-RIS with a pattern designed for the user gives 0.68/1.98/3.15 bit/s/Hz with one-tap for $D = 0.625/0.75/0.875$, 2.98/4.06/5.35 with ours, and 3.08/4.18/5.50 with perfect CSI. A static RIS with phases optimized for the user gives 6.66. Time modulation is therefore not a rate gain for a single user: it is deployed for security, harmonic steering or sensing. Our receiver recovers its cost to within 0.1–0.2 bit/s/Hz of perfect CSI (+2.1–2.3 over one-tap). For security, an eavesdropper using generic CE-BEM already obtains 1.8–2.2 bit/s/Hz, and 2.5–3.0 if it knows the pattern; the pattern must therefore be kept secret.

VI. CONCLUSION

Grouping the harmonic expansion of a dispersive TM-RIS by element state separates the coupling exactly into a part the receiver knows, the switching timing, and a part it learns, the propagation together with the element dispersion. A receiver built on this separation needs no element model. It estimates the coupling from two to eight pilot symbols, finds the RIS clock offset by itself, and equalizes without forming the coupling matrix. In pre-registered tests it beats every baseline we could build on every channel and pilot count, and it decodes within 0.14 dB of perfect CSI. Its limits are stated above: the receiver must know every panel's pattern (A1); delay spreads whose cascade exceeds the CP are out of scope (A2); at 120 km/h on TDL channels pilot overhead and estimate ageing leave 46–66% of the perfect-CSI rate; the stated requirements on jitter, CFO, phase noise and emission apply; and the validation is by simulation with one published cell and no mutual coupling. Over-the-air validation with a real time-modulated surface is the next step.

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